

Williams Cycle effects in Hungarian

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Abstract

A domain exhibits *selective opacity* if it is transparent to some operations but opaque to others. The Williams Cycle (Williams 1974, 2003, 2013) makes a connection between opacity and the *structural size* of embedded clauses: Syntactically smaller embedded clauses are transparent to more sub-extraction types than larger ones. This paper argues that various restrictions on cross-clausal movement in Hungarian can be accounted for in terms of the Williams Cycle. This novel support for the Williams Cycle is welcome since Hungarian is known for exhibiting many movement types landing at various heights (e.g. Horvath 1981; É. Kiss 1987; Puskás 2000; Surányi 2008; Egedi 2021). Bounding can be *overt*: a long movement type landing lower in the matrix left periphery is blocked by constituents occurring high in the embedded left periphery. But bounding can also be *covert*: Even if clause-types (e.g. subjunctive vs. tensed) are not distinguished in the left-periphery, assuming these are of different sizes, their opacity-differences can be traced back to size-differences.

Keywords: Williams Cycle, selective opacity, Hungarian, cross-clausal $\bar{A}$ -movement

1 Introduction

*Selective opacity*¹ emerges when a constituent is opaque to certain syntactic operations but transparent to others. For instance, consider the following tensed clause in English, which can be assumed to be a CP:

¹ The term first appeared in Keine (2016; 2019; 2020).

- (1) [_{CP} *John has drunk a beer.*]

This CP is not an absolute island i.e. it is not opaque to every syntactic operation. For example, *wh*-extraction is possible from this CP:

- (2) *Who*₁ *do you think* [_{CP} ~~*who*~~₁ *has drunk a beer*]?

At the same time, *subject raising* is impossible from CPs as in (3), that is, *hyperraising* is ungrammatical in English (Chomsky 1973).

- (3) **John*₁ *seems* [_{CP} (*that*) ~~*John*~~₁ *has drunk a beer*].

In order to make subject raising possible, we need to replace the CP with a TP:

- (4) *John*₁ *seems* [_{TP} ~~*John*~~₁ *to have drunk a beer*].

One generalization covering this is the *Williams Cycle* (Williams 1974, 2003, 2013) in (5). While subject raising is standardly assumed to land in Spec,TP, *wh*-extraction lands in Spec,CP. Under the Williams Cycle, the ungrammaticality of subject raising from CPs is expected because T is lower than C in the clausal functional sequence.

- (5) **Williams Cycle:** Movement to Spec,βP cannot proceed from Spec,αP or across αP, if α is higher than β in some functional sequence.

(based on Williams 2003; Poole 2023)

In general, the Williams Cycle derives opacity from *structural size* (rather than other factors like the A vs. $\bar{A}$ distinction): Larger clauses are opaque to more types of subextraction.

In this article, I demonstrate that Hungarian exhibits Williams Cycle effects. Here I handle the Williams Cycle as a descriptive generalization and I do not provide a theoretical explanation why the Williams Cycle holds. Roughly any account capable of deriving the Williams Cycle (e.g. Williams 2003, 2013; Abels 2012; Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2023; Egressy 2023) can be applied to the Hungarian data.

In some cases, clause size is indicated by the presence of some left-periphery constituent; in these cases, we are dealing with *overtly driven bounding*.² For instance, if the embedded clause has a focused constituent as in (6), we know that it is at least a FocP, and hence movement types landing below the matrix FocP are predicted to be ungrammatical from within this FocP. Thus, assuming Foc>Part in the functional sequence, particle movement landing in Spec,PartP is correctly predicted to be ungrammatical from the FocP in (6).

- (6)
$$\begin{array}{c} \downarrow \text{-----} \uparrow \\ [\text{PartP} \times \dots [\text{VP} \dots [\text{FocP} \text{ focus } [\dots \text{XP} \dots]]]] \end{array}$$
- $[\text{PART} \text{ Keresztül}]_1 \text{ akar-}t\text{-}am, \quad [\text{hogy } (*[\text{FOC} \text{ CSAK } \text{MARI-VAL}])$
across want-PST-1SG.DEF that only Mari-INS
gázol-j-anak a patak-on $__1]$
wade-SUBJ-3PL.INDF the brook-SUP
‘I wanted them to wade across the brook (*ONLY WITH MARI).’

However, it is not necessarily the case that clause-size differences are indicated by overt material in the left-periphery. For instance, the difference between subjunctive clauses and tensed clauses is not indicated in the left-periphery. Thus, even if particle movement was possible from a subjunctive clause (6), once the embedded clause is tensed as in (7), particle movement becomes impossible. If the impossibility of particle movement from tensed clauses is derived from size (i.e. tensed clauses are larger than subjunctive clauses), this can be categorized as *covertly driven bounding*.

- (7)
$$\begin{array}{c} \downarrow \text{-----} \uparrow \\ [\text{PartP} \times \dots [\text{VP} \text{ lát } [\dots \text{V}_{\text{Tensed}} [\dots \text{XP} \dots]]]] \end{array}$$
- $^{??}[\text{PART} \text{ Keresztül}]_1 \text{ lát-}t\text{-}am, \quad [_{\text{Tensed}} \text{hogy } __1 \text{gázol-}t\text{-}ak \quad a$
across see-PST-1SG.DEF that wade-PST-3PL.INDF the
patak-on $__1]$
brook-SUP
Intended: ‘I saw that they waded across the brook.’

² Throughout the article, the term *bounding* is generally used for "blocking movement and agreement by dominating categories". This general definition does not limit *bounding* to concepts like *bounding nodes*, *blocking categories*, or *barriers* (as it happens in works like Chomsky 1986).

Finally, not all types of overt material indicate clause-size differences: For example, even though topics and quantifiers can occur around the linear position of sentential adverbs and foci in the left periphery, among these, only sentential adverbs and foci lead to bounding effects. This does not disturb the Williams Cycle claims for Hungarian assuming only relative pronouns, sentential adverbs, foci, and particles indicate clause-size differences by occurring in the specifier of some dedicated projection, while topics and quantifiers can be handled as adjuncts to clauses of various sizes (as in Szendrői 2003; É. Kiss 2010). Thus, beyond demonstrating that Hungarian exhibits Williams Cycle-style bounding in general, the main contribution of this article is that it shows that not every instance of size- and opacity-difference is overtly expressed in the left-periphery and some kinds of left-periphery material do not contribute to bounding.

The discussion proceeds as follows: In Section 2, I introduce my assumptions on the Hungarian left-periphery, and I present the Hungarian long-movement data. On the basis of these, I claim that Hungarian exhibits Williams Cycle effects. Section 3 discusses topics and quantifiers as apparent counterexamples. Section 4 adds the tensed vs. subjunctive clause-type difference. The article is concluded by Section 5, where I also discuss a potential way to extend a Williams Cycle-based approach to Hungarian definiteness agreement.

2 Overtly driven bounding as Williams Cycle

This section introduces the relevant positions in the Hungarian left-periphery and demonstrates how the presence of these left-periphery positions influences the opacity of embedded clauses. The opacity of clauses will be exemplified by the possibility of three movement types: particle-movement, focus-movement, and relative movement. After assigning projections to each of the left-periphery positions, I propose that the opacity-patterns of the various movement types align with a strict version of the Williams Cycle (Williams 1974, 2003, 2013).

Before discussing the various movement types, the sentence in (8) provides an overview of the clause structure. In this sentence, all left-periphery positions are represented: The

relative pronoun is followed by a high sentential adverb, which is followed by a focus, which is followed by a verbal particle. I assume that the left-periphery order reflects the hierarchical relation between the phrases hosting these constituents: Rel(ative)P hosting relative pronouns is the highest projection in the clause. This article uses the term *sentential adverbs* for adverbs expressing the speaker’s evaluation of the proposition (e.g. *szerencsére* ‘fortunately’) or connecting the sentences of the discourse (e.g. *következésképpen* ‘consequently’). These adverbs do not modify the truth conditional content of the proposition but specify the speaker’s attitude towards the proposition and locate the sentences in the discourse. The presence of such adverbs indicates that the embedded clause denotes the content of some speech act. That we are dealing with an embedded clause of this kind is encoded by the Com(mitment)P layer in the syntactic structure (the projection’s name comes from Krifka 2023). ComP is assumed to host sentential adverbs in its specifier.³ ComP dominates Foc(us)P hosting foci, which dominates Part(icle)P hosting particles. Why topics and quantifiers do not get dedicated projections in the functional sequence is discussed in Section 3.⁴

³ Krifka (2023) introduces the layers JudgeP, Com(mitment)P, and ActP to host various adverbs specifying speaker-attitude towards the proposition and its rhetorical status. In this paper, I do not discuss adverbs of the same form occurring in the right-periphery. Even if we handle sentential adverbs as adjuncts, my central claim does not change: Their presence indicates that the clause is of a certain size (in this paper, ComP). I assume that ComP may be projected even if no overt sentence adverb occupies its specifier.

⁴ Although I am a native speaker of Hungarian, the grammaticality of critical sentence types were double-checked in two informal surveys. The first one was conducted in Summer, 2022, the second one in Spring, 2023. The surveys had 15 and 35 native-speaker participants, respectively.

In the bracket diagrams occurring above Hungarian examples, embedded clauses are highlighted in gray, dashed arrows refer to sub-extraction from embedded clauses, and simple arrows refer to all other types of movement. In first line of the glossing, the various syntactic positions (focus, particle, relative etc.) are marked with subscripts. These subscripts do not specify whether a constituent is an adjunct or specifier. The glossing follows the Leipzig Glossing rules. The abbreviations are the following: 1 = first person, 2 = second person, 3 = third person, ACC = accusative, COND = conditional, DEF = definite, FOC = focus, IN = inessive, INDF = indefinite, INF = infinitive, INS = instrumental, PART = particle, PL = plural, POT = potential, PST = past, Q = quantifier, REL = relative, S.Adv = sentential adverb, SG = singular, SUB = sublativ, SUBJ = subjunctive, SUP = superessive, TOP = topic

pre-verbal position (É. Kiss 1998b; Dékány and Hegedűs 2015), which I will simply call Spec,PartP. This is shown in the following sentence: ⁵

- (11) $\downarrow$
 $[_{\text{PartP}} \checkmark [_{\text{VP}} [_{\text{PP}} \dots \text{XP} \dots]]]$
- $[_{\text{PART}} \text{Keresztül}]_1 \text{ gázol-t-am} \quad [_{\text{PP}} \text{a} \text{ patak-on} \text{ } __1]$
 across wade-PST-1SG.INDF the brook-SUP
 ‘I waded across the brook.’

Particle movement can proceed across clauses as well. In (12), particle movement proceeds from the complement clause of *akar* ‘want’:

- (12) $\downarrow$
 $[_{\text{PartP}} \checkmark \dots [_{\text{CP}} \dots [_{\text{VP}} [_{\text{PP}} \dots \text{XP} \dots]]]]$
- $[_{\text{PART}} \text{Keresztül}]_1 \text{ akar-t-am}, \quad [\text{hogy} \text{ gázol-j-anak} \quad \text{a} \text{ patak-on} \text{ } __1]$
 across want-PST-1SG.DEF that wade-SUBJ-3PL.INDF the brook-SUP
 ‘I wanted them to wade across the brook.’

Particle movement is blocked by various kinds of intervening material in the embedded left-periphery. First of all, if the embedded Spec,PartP is filled by another particle, long-particle movement is blocked:

- (13) $\downarrow$
 $[_{\text{PartP}} \times \dots [_{\text{VP}} \dots [_{\text{PartP}} \text{particle} [\dots \text{XP} \dots]]]]$
- $[_{\text{PART}} \text{Keresztül}]_1 \text{ akar-t-am}, \quad [\text{hogy} \text{ } (*[_{\text{PART}} \text{át}]) \text{ gázol-j-anak} \text{ } __1]$
 across want-PST-1SG.DEF that through wade-SUBJ-3PL.INDF
 a patak-on
 the brook-SUP
 ‘I wanted them to wade across the brook (through).’

⁵ In the literature this has been called Spec,AspP (É. Kiss 2002), Spec,PredP (Csirmaz 2004), as well as Spec,TP (Olsvay 2000; Surányi 2009, 2012).

In a similar way, long-particle movement is also blocked by foci occurring in the embedded clause's Spec,FocP position (Farkas and Sadock 1989):

- (14) $\downarrow$ -----
 [PartP $\times$... [VP ... [FocP focus [... XP ...]]]]
 [PART *Keresztül*]₁ *akar-t-am*, [*hogy* (*[FOC CSAK MARI-VAL])
 across want-PST-1SG.DEF that only Mari-INS
gázol-j-anak *a patak-on* ___₁]
 wade-SUBJ-3PL.INDF the brook-SUP
 'I wanted them to wade across the brook (*ONLY WITH MARI).'

Particle movement is blocked by intervening high sentential adverbs like *szerencsére* 'fortunately' and *következésképpen* 'consequently' in the embedded clause as in (15). Again, I assume that all high sentential adverbs are hosted in Spec,Com(mitment)P (label from Krifka 2023).

- (15) $\downarrow$ -----
 [PartP $\times$... [VP ... [Comp sentential adverbs [... XP ...]]]]
 [PART *Keresztül*]₁ *akar-t-am*, [*hogy* ([EVAL *következésképpen])
 across want-PST-1SG.DEF that consequently
gázol-j-anak *a patak-on* ___₁]
 wade-SUBJ-3PL.INDF the brook-SUP
 'I wanted them to wade (consequently) across the brook.'

The second movement type to be covered is focus movement, which lands in front of particles in Spec,FocP, and triggers the movement of the verb to Foc, which is a focus-adjacent position as in (16) (Horvath 1986; É. Kiss 1987).

- (16) $\downarrow$ -----
 [FocP $\checkmark$ Foc+V[PartP[VP V ... XP ...]]]
 [FOC (CSAK) A PATAK-ON]₂ *gázol-t-am*₁ [PART *keresztül*]₃ ___₁ ___₂ ___₃
 only the brook-SUP wade-PST-1SG.INDF across
 'I have waded across ONLY THE BROOK.'

(17)
$$[\text{FocP } \checkmark \text{ Foc+V}[\text{PartP}[\text{VP } \text{V} \dots \text{wh-XP} \dots]]]$$

$$[\text{FOC } \textit{Melyik } \textit{patak-on}]_2 \textit{ gázol-t-ál}_1 \quad [\text{FOC } \textit{keresztül}]_3 \text{ } __1 \text{ } __2 \text{ } __3?$$

which brook-SUP wade-PST-2SG.INDF across

‘Which brook did you wade across?’

(18) $[\text{FocP } \checkmark \dots [\text{CP } \dots [\text{VP} [\text{PP } \dots \text{XP } \dots]]]]$

$[\text{FOC } \text{CSAK } \text{A } \text{PATAK-ON}]_2 \text{ akar-t-am}_1 \quad \text{---}_1, [\text{ hogy } [\text{PART } \text{keresztül}]_3$

only the brook-SUP want-PST-1SG.DEF that across

$\text{gázol-j-anak} \quad \text{---}_2 \text{ ---}_3]$.

wade-SUBJ-3PL.INDF

‘I wanted them to wade across ONLY THE BROOK.’

(19) $[\text{FocP } \checkmark \dots [\text{VP } \dots [\text{FocP } \text{focus } [\dots \text{XP } \dots]]]]$

$[\text{FOC } \text{CSAK } \text{A } \text{PATAK-ON}_1] \text{ akar-}t\text{-am}_2 \quad \text{---}_2, [\text{hogy } [\text{FOC } \text{CSAK } \text{MARI-VAL}]$
 only the brook-SUP want-PST-1SG.DEF that only Mari-INS
 $\text{gázol-j-anak}_3 \quad [\text{PART } \text{keresztül}]_4 \text{ ---}_3 \text{ ---}_1 \text{ ---}_4]$
 wade-SUBJ-3PL.INDF across
 ‘ONLY THE BROOK I wanted them to wade across ONLY WITH MARI.’

9

- (20) $\downarrow$ -----
 [FocP $\times$... [VP ... [Comp sentential adverbs [... XP ...]]]]
 [FOC (CSAK) A PATAK-ON]₂ hall-ott-am₁ ____₁, [hogy (??[_{eval} szerencsére])
 only the brook-SUP hear-PST-1SG.DEF that fortunately
 [_{part} keresztül]₃ gázol-t-ak ____₂ ____₃].
 across wade-PST-3PL.INDF
 ‘I heard that (*fortunately) they waded across ONLY THE BROOK.’

The last movement-type is the movement of relative pronouns. This movement lands in Spec,RelP, in front of sentential adverbs, foci, and particles:⁶

- (21) $\downarrow$ -----
 [RelP $\checkmark$ [Comp[FocP[PartP [VP[PP ... XP ...]]]]]]
 Ez az a patak, [_{eval} *következésképpen/ *szerencsére] [_{rel} amelyik-en]
 this that the brook, consequently/ fortunately which-SUP
 [_{eval} következésképpen/ szerencsére] [_{foc} CSAK PETI-VEL]₂ gázol-t-am₃
 consequently/ fortunately only Peti-INS wade-PST-1SG.INDF
 ____₂ [_{part} keresztül]₄ ____₃ ____₁ ____₄
 across
 ‘This is the brook, which consequently/fortunately I waded across ONLY WITH PETI.’

Long relative movement can cross embedded particles as in (22), foci as in (23), and high sentential adverbs as in (24):

- (22) $\downarrow$ -----
 [RelP $\checkmark$... [VP ... [PartP particle [... XP ...]]]]
 Ez az a patak, [_{rel} amelyik-en]₁ hall-ott-am, [hogy [_{part} keresztül]₃
 this is the brook which-SUP hear-PST-1SG.DEF that across
 gázol-t-ak ____₁ ____₃]
 wade-PST-3PL.INDF
 ‘This is the brook which I heard that they waded across.’

⁶ Other analyses say that relative movement targets Spec,ForceP (Rizzi 1997). This does not make any difference as long as the landing site is higher than all other projections discussed here.

- (23) $\downarrow$ $\text{[RelP } \checkmark \dots \text{[VP } \dots \text{[FocP focus [} \dots \text{XP } \dots \text{]}]]]}$
- Ez az a patak, [REL amelyik-en]₁ az-t akar-t-am, [hogy [FOC*
 this that the brook which-SUP it-ACC want-PST-1SG.DEF that
CSAK MARI-VAL] gázol-j-anak₂ [PART keresztül]₃ ___₂ ___₁ ___₃]
 only Mari-INS wade-SUBJ-3PL.INDF across
 ‘This is the brook which I wanted them to wade across ONLY WITH MARI.’

- (24) $\downarrow$ $\text{[RelP } \checkmark \dots \text{[VP } \dots \text{[ComP sentential adverbs [} \dots \text{XP } \dots \text{]}]]}$
- Ez az a patak, [REL amelyik-en]₁ hall-ott-am, [hogy [EVAL szerencsére]*
 this that the brook which-SUP hear-PST-1SG.DEF that fortunately
[PART keresztül]₂ gázol-t-ak ___₁ ___₂]
 across wade-PST-3PL.INDF
 ‘This is the brook which I heard that they fortunately waded across.’

The intervention-sensitivity of the various movement types is shown again in the table below:

(25) **How does intervening left-periphery material influence opacity?**

	Particle extraction	Focus extraction	Relative extraction
Filled Spec,PartP	×	✓	✓
Filled Spec,FocP	×	✓	✓
Filled Spec,ComP	×	×	✓

Again, I assume the following *fseq* for the Hungarian left-periphery:

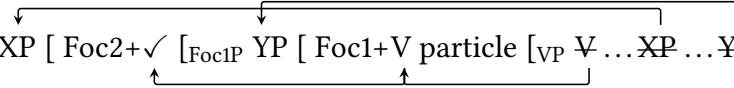
- (26) Rel>Com>Foc>Part

On the basis of (25) and (26), we can draw the following generalization:

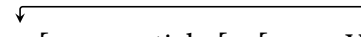
- (27) Movement to Spec, β P cannot proceed across a filled Spec, α P, where $\alpha \geq \beta$ in the functional sequence.

In other words, once XP crossed a Spec, α P left-periphery position in any clause, it can only land in the specifier of phrases whose head is higher than α in the functional sequence.

There seems to be an exception to this generalization: Foci can move across foci. This is explained by É.Kiss's (1998a) observation that foci can be stacked within clauses in Hungarian:

- (28) 
- [_{Foc2} CSAK KÉT LÁNY]₁ olvas-ott-Ø₂ [_{Foc1} CSAK EGY KÖNYV-ET]₃ ₂ [_{PART}
 only two girl read-PST-3SG.INDF only one book-ACC
 el]₂ ₁ ₃ a vizsgá-ra
 across the exam-SUB
 ‘ONLY TWO GIRLS were such that they read ONLY ONE BOOK for the exam.’

For sentences like (28), É.Kiss (1998a) proposes two focus projections: Foc1P and Foc2P. If the Foc1P projection headed by Foc1 is distinct from the higher projection Foc2P headed by Foc2, it is explained why the verb moves to a position between the two foci: V first moves to Foc1 then to Foc2. Notice that no such stacking is possible with particles:

- (29) 
- [_{PART} Keresztül]₁ (*[_{PART} át]) gázol-t-ak a patak-on ₁
 across through wade-PST-3PL.INDF the brook-SUP
 Intended: ‘They waded away across the brook.’

Given that we have two focus projections such that Foc2P > Foc1P, focus movement from clauses with foci like (19) can be re-analyzed as focus movement to the Spec,Foc2P of the matrix clause across a filled embedded Spec,Foc1P, which does not violate (27). In short, the generalization in (27) can be maintained. For the sake of simplicity, I do not use multiple focus projections in the rest of the paper.

The generalization on overtly driven bounding in (27) is reminiscent of the *Williams Cycle* (Williams 1974, 2003). The Williams Cycle is usually formulated as follows:

- (30) **Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, if α is higher than β in some functional sequence.
(based on Williams 2003; Poole 2023)

The Williams Cycle has been shown to hold in English (Williams 2003); its most typical illustration is the ban on *hyperraising* in English: According to the Williams Cycle, the sentence in (31) is ungrammatical because raising to subject, which lands in matrix Spec,TP, crosses the embedded CP, and C is higher than T in the functional sequence.

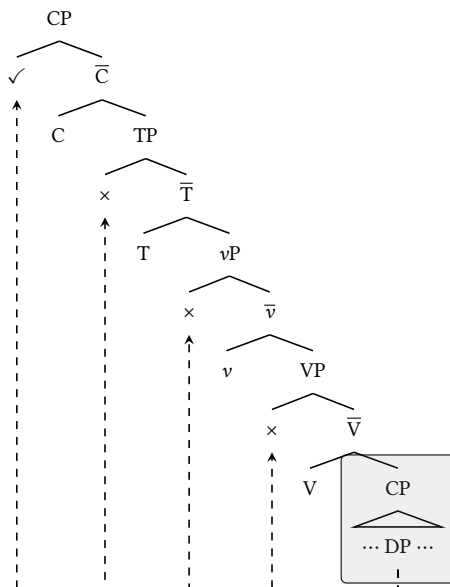
- (31) * $[_{TP} John_1 \text{ seems } [_{CP} (that) \text{ } \cancel{John}_1 \text{ has drunk a beer}]]$

Notice that no such constraint holds in the case of raising from within TPs because no crossed head is higher in *fseq* than the lastly crossed matrix T head.

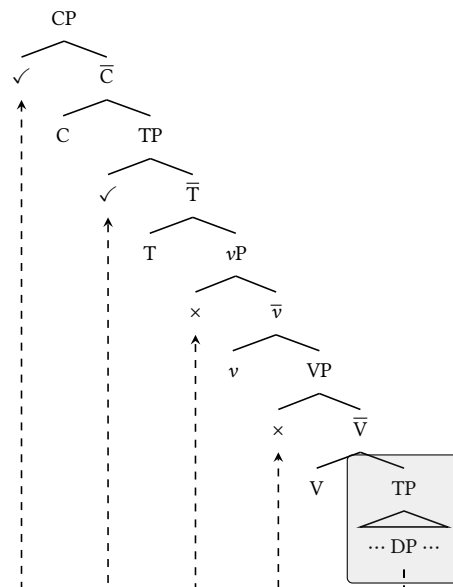
- (32) $[_{TP} John_1 \text{ seems } [_{TP} \cancel{John}_1 \text{ to have drunk a beer}]]$

The difference in opacity between English CPs and TPs can be depicted by the structures in (33) and (34), respectively. Movement from within CPs can only land in matrix Spec,CP, whereas movement from within TPs can land in Spec,TP or Spec,CP but not in Spec,vP or Spec,VP.

- (33) The opacity of English CPs



- (34) The opacity of English TPs



While the better known constraint, the Ban on Improper Movement (Chomsky 1973),

covers some of the ground covered by the Williams Cycle, e.g. movement to Spec,CP, an $\bar{A}$ position, cannot be followed by movement to Spec,TP, an A position, the Williams Cycle has predictions beyond the A vs. $\bar{A}$ distinction. This is essential because the Hungarian left-periphery positions discussed in the paper are all $\bar{A}$ -positions.

In order to align the generalization in (27) and the Williams Cycle in (30), three adjustments are needed. Firstly, I assume that the complementizer *hogy* ‘that’ is not a head in the clausal *fseq*, that is, there is no CP projection in the Hungarian clausal spine. Instead, *hogy* is simply an adjunct, an edge marker without a dedicated projection along the clausal spine (cf. Manetta 2006, 2011; Poole 2023). That is, embedded clauses are not uniformly CPs but PartPs, FocPs, ComPs, and RelPs. This idea is not entirely new: Koopman and Szabolcsi (2000) also proposed that *hogy* is not generated as a projecting head in the embedded clause. Their main argument is that *hogy* can introduce a variety of clause types: Subjunctives as well as tensed clauses (cf. Section 3). In addition, *hogy* does not have to occur with clauses with propositional content but it can stand in front of sounds imitating someone’s interjection:

- (35) *Sólyomszem az-t mond-t-a, [hogy nyá]*
Hawkeye it-ACC say-PST-3SG.DEF that gee
‘Hawkeye said (*that) gee.’

The translation of (35) indicates that English *that* is not compatible with sounds of reaction. I will handle *hogy* as an adjunct to the embedded clause.

Secondly, once embedded clauses are not uniformly CPs, it needs to be made explicit that if a clause has an operator in its Spec, α P, it is at least an α P. For instance, embedded clauses with a focus will be FocPs and embedded clauses with high adverbs will be ComPs. Consequently, while matrix verbs require a minimal size for their complements (e.g. PartP, TP, MoodP, or ComP), their complement clauses can grow larger (e.g. to FocP, ComP, RelP) (for a similar idea, see Deal 2020). It is not central here to decide whether projections can be missing from the middle of extended projections, that is, whether the presence of higher projections requires the presence of lower ones (for some discussion on the matter, see e.g. Ramchand and Svenonius 2014).

The third adjustment concerns the tolerance of crossing of phrases of the same kind. In English, movement from embedded CPs to matrix Spec,CPs and from embedded Spec,TPs to matrix Spec,TPs is not ruled out. The Williams Cycle in (30) allows such movement. On the other hand, the movement of particles and relative pronouns cannot cross particles and relative pronouns, respectively.⁷ In order to account for the Hungarian data, I propose a *strict* version for the Williams Cycle:⁸

- (36) **Strict Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, if $\alpha \geq \beta$ in some functional sequence.

That is, movement from within PartP to matrix Spec,PartP is ruled out just as movement from within RelP to matrix Spec,RelP. Again, the possibility of focus movement across other foci is accounted for by multiple focus projections. For structures like (28), I assume that Foc2 is α and Foc1 is β . Thus, the movement from the embedded clause can proceed across Foc1P and land in Foc2P without the violation of (36) if Foc1 > Foc2.

How widely the strict version in (36) holds remains an open question. The strict version rules out the widely attested patterns of successive cyclic movement from CP to Spec,CP and TP to Spec,TP (for a survey, see Georgi 2014, 2017), which the original version in (30) allowed. At the same time, Italian focus extraction across foci is ruled out (Abels 2012), and Zulu TPs are opaque to raising to matrix Spec,TP (Halpert 2019). In other words, even if (36) cannot universally replace (30), the Hungarian patterns discussed here are not isolated phenomena.

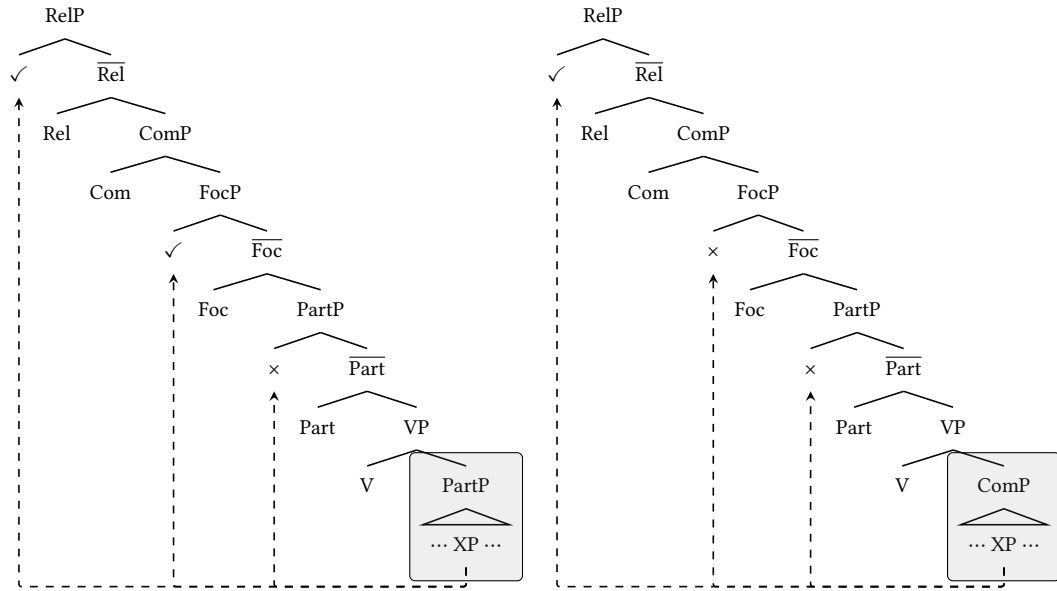
After these adjustments, the (strict) Williams Cycle in Hungarian can be exemplified by embedded PartPs and ComPs in (37) and (38), respectively. Since Com > Part in *fseq*,

⁷ For the ban on crossing relative pronouns, one could ask why reference to the Complex NP Constraint is not sufficient. In this paper, I follow a different attitude by saying that the Complex NP constraint follows from a more general constraint, the (strict) Williams Cycle. In this spirit, relative clauses are not islands in an absolute sense, but most languages lack the movement type that lands high enough to escape them. Relative clauses in languages like Swedish (Allwood 1982) are not absolute islands to *wh*-movement. With a similar attitude, Starke (2001) argues that even strong islands can be handled in a unified framework of selective opacity.

⁸ The theoretical accounts mentioned in the introduction can also derive the strict Williams Cycle in (36) after minor adjustments. An alternative to proposing a stricter Williams Cycle is to introduce special conditions for particle and relative movement (cf. Egressy 2023). The most obvious candidate would be to propose Relativized Minimality (Rizzi 1990) and say that if there is a higher particle or relative pronoun, the lower particle or relative pronoun cannot undergo movement.

every movement type that is possible from ComP is also possible from PartP.⁹

- (37) The opacity of Hungarian PartPs (38) The opacity of Hungarian ComPs



3 Apparent counterexamples: Topics and quantifiers

Topics and quantifiers have not been discussed so far. Similar to sentential adverbs, topics and quantifiers *can* occur in front of particles and foci and behind relative pronouns:

- (39) $[_{RelP} \text{ relative pronouns } [_{\{ \text{topics/ quantifiers} \}} [_{FocP} \text{ foci } [_{PartP} \text{ particles } [VP]]]]]$
A patak, [REL amelyik-en]₁ { [TOP vasárnap]/ [Q minden nap] } [FOC CSAK]
 the brook which-SUP Sunday every day only
PETI-VEL] gázol-t-am₂ [PART keresztül]₃ ___₂ ___₁ ___₃
 Peti-INS wade-PST-1SG.INDF across
 ‘The brook, which {on Sunday/ every day} I waded across ONLY WITH PETI.’

On the basis of the linear order in (39), one may expect that the presence of topics and quantifiers have the same effects on bounding as the presence of sentential adverbs.

⁹ One account capable of handling the instances of overtly driven bounding as intervention (in the sense of Relativized Minimality) is that of Abels (2012). Since this theory does not account naturally for the clause type data presented in the following sections, I do not discuss it in this paper.

That is, just as sentential adverbs preceding particles and foci within clauses could not be crossed by long moving particles and foci, we expect that topics and quantifiers capable of occurring roughly in the region of high sentential adverbs block the long movement of particles and foci. However, this does seem to be the case. Topics do not block the long movement of particles as in (40) (Koopman and Szabolcsi 2000), foci as in (41), and relative pronouns (42).

- (40) $\downarrow$ -----
 [PartP ✓ ... [VP ... [topic [... XP ...]]]]
 [_{PART} *Keresztül*]₁ *akar-t-am*, [*hogy* [_{TOP} *vasárnap*] *gázol-j-anak*
 across want-PST-1SG.DEF that vasárnap wade-SUBJ-3PL.INDF
a patak-on ____₁]
 the brook-SUP
 ‘I wanted them to wade across the brook on Sunday.’

- (41) $\downarrow$ -----
 [FocP ✓ ... [VP ... [topic [... XP ...]]]]
 [_{FOC} (*CSAK*) *A PATAK-ON*]₂ *hall-ott-am*₁ ____₁, [*hogy* [_{TOP} *vasárnap*] [_{PART}
 only the brook-SUP hear-PST-1SG.DEF that Sunday
keresztül]₃ *gázol-t-ak* ____₂ ____₃].
 across wade-PST-3PL.INDF
 ‘I heard that on Sunday, they waded across ONLY THE BROOK.’

- (42) $\downarrow$ -----
 [RelP ✓ ... [VP ... [topic [... XP ...]]]]
Ez az a patak, [_{REL} *amelyik-en*]₁ *hall-ott-am*, [*hogy* [_{TOP} *vasárnap*]
 this that the brook which-SUP hear-PST-1SG.DEF that Sunday
 [_{PART} *keresztül*]₂ *gázol-t-ak* ____₁ ____₂]
 across wade-PST-3PL.INDF
 ‘This is the brook which I heard that on Sunday, they waded across.’

In a similar way, quantifiers do not block the sub-extraction of particles as in (43) (Koopman and Szabolcsi 2000), foci as in (44), and relative pronouns (45).

- (43) $\begin{array}{c} \downarrow \\ \text{[PartP } \checkmark \dots [\text{VP } \dots [\text{quantifier [} \dots \text{ XP } \dots]]]] \end{array}$
- $\text{[PART Keresztül]}_1 \text{ akar-t-am, } [\text{hogy [}_Q \text{ minden nap]} \text{ gázol-j-anak}$
 across want-PST-1SG.DEF that every day wade-SUBJ-3PL.INDF
 $\text{a patak-on } __1]$
 the brook-SUP
 ‘I wanted them to wade across the brook every day.’

- (44) $\begin{array}{c} \downarrow \\ \text{[FocP } \checkmark \dots [\text{VP } \dots [\text{quantifier [} \dots \text{ XP } \dots]]]] \end{array}$
- $\text{[FOC (CSAK) A PATAK-ON]}_2 \text{ hall-ott-am}_1 __1, [\text{hogy [}_Q \text{ minden nap]} \text{ [PART}$
 only the brook-SUP hear-PST-1SG.DEF that every day
 $\text{keresztül]}_3 \text{ gázol-t-ak } __2 __3].$
 across wade-PST-3PL.INDF
 ‘I heard that they waded across ONLY THE BROOK every day.’

- (45) $\begin{array}{c} \downarrow \\ \text{[RelP } \checkmark \dots [\text{VP } \dots [\text{quantifier [} \dots \text{ XP } \dots]]]] \end{array}$
- $\text{Ez az a patak, [REL amelyik-en]}_1 \text{ hall-ott-am, } [\text{hogy [}_Q \text{ minden nap]}]$
 this that the brook which-SUP hear-PST-1SG.DEF that every day
 $\text{[PART keresztül]}_2 \text{ gázol-t-ak } __1 __2]$
 across wade-PST-3PL.INDF
 ‘This is the brook which I heard that every day they waded across.’

At the same time, the Williams Cycle in (36) does not make reference to linear positions in clauses but to extended projections and functional sequences. That is, we expect blocking effects from left-periphery constituents which are actually indicative of clause size. For example, topics and quantifiers could only have the same blocking effects as sentential adverbs, if the functional sequence had a dedicated Top(ic) head and a dedicated Q(uantifier) head, which would have approximately the following position in the hierarchy: Rel>Top>Com>Q>Foc>Part>V.

Here I follow Surányi (2002) and É.Kiss (2010) in assuming that Hungarian quantifiers are adjuncts without dedicated projections in the *fseq* and Szendrői (2003) in assuming

that Hungarian topics are adjuncts without dedicated projections in the *fseq*, too. Instead, quantifiers and topics are assumed to adjoin to phrases of various sizes. The most extreme example of this is that unlike foci, topics and quantifiers can occur in radically truncated clauses lacking external arguments and accusative case marking as in (46) (Halm 2021):

(46) [{ topics/quantifiers/*foci} [VP [...]]]

{_{TOP} *Bejárat elött*}/ [_Q *minden bejárat elött*]/ *_{FOC} *CSAK BEJÁRAT*
 entrance in.front.of every entrance in.front.of only entrance
ELÖTT] *láb meg töröl*
 in.front.of foot PRT wipe
 ‘{ In front of the entrance/ in front of every entrance/ *ONLY IN FRONT OF THE
 ENTRANCE}, (someone) wipes their feet (shoes).’

These presumably VP-sized clauses do not have a Spec,FocP position for foci, but topics and quantifiers can adjoin to their highest projection.

In sum, topics and quantifiers are irrelevant to Williams Cycle analysis of the Hungarian data because they do not have dedicated projections in the clausal extended projection. Whether topics and quantifiers are involved in Williams Cycle effects in other languages remains an open question.

4 Covertly driven bounding: Subjunctive vs. tensed clauses

So far, I have not discussed the clause types defined by the verb heading the embedded clauses. In this section, I show that although clause types do not differ in their overt left periphery, opacity depends on clause type just as it depended on the amount and type of overt left-periphery material. In order to provide a unified analysis of selective opacity, I derive the difference between clause-types from syntactic size.

Hungarian matrix verbs select for various kinds of embedded clauses. For example, *akar* ‘want’ selects for clauses headed by subjunctive verbs, while *lát* ‘see’ selects for clauses headed by tensed verbs:

- (47) *Akar-t-ad,* [Subjunctive *hogy Péter {hasít-s-a/ *hasít-ott-a}* *fel*
 want-PST-2SG.DEF that Péter cut-SUBJ-3SG.DEF cut-PST-3SG.DEF up
a fá-t]
 the wood-ACC
 ‘You wanted Péter to cut the wood.’

- (48) *Lát-t-ad,* [Tensed *hogy Péter {hasít-ott-a/ *hasít-s-a}* *fel a*
 see-PST-2SG.DEF that Péter cut-PST-3SG.DEF cut-SUBJ-3SG.DEF up the
fá-t]
 wood-ACC
 ‘You saw that Péter was cutting the wood.’

While long particle extraction is possible from the complement clause of *akar* ‘want’, it is impossible from the complement clause of *lát* ‘see’ (Farkas and Sadock 1989; É. Kiss 1994; Koopman and Szabolcsi 2000):¹⁰

- (49) [PartP ✓ ... [VP *akar* [... V_{Subjunctive} [... XP ...]]]
 [PART *Keresztül*]₁ *akar-t-am,* [Subj *hogy gázol-j-anak a patak-on*
 across want-PST-1SG.DEF that wade-SUBJ-3PL.INDF the brook-SUP
 —₁]

‘I wanted them to wade across the brook.’

¹⁰ The literature usually remains implicit about the impossibility of particle climbing from tensed clauses. The generalization was most explicitly formed by É.Kiss (1994), who says that the particles’ “long movement, however, is more constrained with respect to locality than the long movement of a full XP projection: while full XP projections can, in general, be extracted out of governed infinitival and non-indicative clauses, in the case of the long extraction of a verbal prefix it is also required that the source clause be governed by a modal matrix V” (É. Kiss 1994:p. 44). I infer that the opacity of tensed clauses has been known but considered trivial by the authors of the studies.

- (50) $\downarrow$ $\text{[PartP} \times \dots \text{[VP} \textit{lát} \text{ [} \dots \text{V}_{\text{Tensed}} \text{ [} \dots \text{XP} \dots \text{]]}]}$
- * $\text{[}_{\text{PART}} \textit{Keresztül}]_1 \textit{lát-t-am}, \quad \text{[}_{\text{Tensed}} \textit{hogy} \text{ } __1 \textit{gázol-t-ak} \quad \textit{a}$
 across see-PST-1SG.DEF that wade-PST-3PL.INDF the
 $\textit{patak-on} \text{ } __1]$
 brook-SUP
 Intended: ‘I saw that they waded across the brook.’

Focus movement and relative movement are possible from both subjunctive and tensed clauses:

- (51) $\downarrow$ $\text{[FocP} \checkmark \dots \text{[VP} \textit{akar} \text{ [} \dots \text{V}_{\text{Subjunctive}} \text{ [} \dots \text{XP} \dots \text{]]}]}$
- $\text{[}_{\text{FOC}} \textit{Melyik} \textit{patak-on}]_1 \textit{akar-t-ad}_2 \quad __2, \text{[}_{\text{Subj}} \textit{hogy} \text{ [}_{\text{PART}} \textit{keresztül}]_3$
 which brook-SUP want-PST-2SG.DEF that across
 $\textit{gázol-j-anak} \quad __1 \text{ } __3]$?
 wade-SUBJ-3PL.INDF
 ‘Which brook did you want them to wade across?’

- (52) $\downarrow$ $\text{[FocP} \checkmark \dots \text{[VP} \textit{lát} \text{ [} \dots \text{V}_{\text{Tensed}} \text{ [} \dots \text{XP} \dots \text{]]}]}$
- $\text{[}_{\text{FOC}} \textit{Melyik} \textit{patak-on}]_1 \textit{lát-t-ad}_2 \quad __2, \text{[}_{\text{Tensed}} \textit{hogy} \text{ [}_{\text{PART}} \textit{keresztül}]_3$
 which brook-SUP see-PST-2SG.DEF that across
 $\textit{gázol-t-ak} \quad __1 \text{ } __3]$?
 wade-PST-3PL.INDF
 ‘Which brook did you see that they waded across?’

- (53) $\downarrow$ $\text{[RelP} \checkmark \dots \text{[VP} \textit{akar} \text{ [} \dots \text{V}_{\text{Subjunctive}} \text{ [} \dots \text{XP} \dots \text{]]}]}$
- $\textit{Ez} \textit{az} \textit{a} \textit{patak}, \text{[}_{\text{REL}} \textit{amelyik-en}]_1 \textit{az-t} \textit{akar-t-ad}, \quad \text{[}_{\text{Subj}} \textit{hogy} \text{ [}_{\text{PART}}$
 this that the brook which-SUP it-ACC want-PST-2SG.DEF that
 $\textit{keresztül}]_2 \textit{gázol-j-anak} \quad __1 \text{ } __2]$
 across wade-SUBJ-3PL.INDF
 ‘This is the brook which you wanted them to wade across.’

- (54) $\downarrow$ -----
 [RelP $\checkmark$... [VP lát [... V_{Tensed} [... XP ...]]]
Ez az a patak, [REL amelyik-en]₁ lát-t-ad, [Tensed hogy [PART
 this is the brook which-SUP see-PST-1PL.DEF that
keresztül]₂ gázol-t-ak $\text{---}_1 \text{---}_2$
 across wade-PST-1PL.INDF
 ‘This is the brook which you saw that they waded across.’

Unlike certain Indo-European languages (e.g. Latin or German), Hungarian subjunctives do not make a tense-distinction. As in many Balkan languages, subjunctive and infinitival clauses have a very similar role in Hungarian grammar.¹¹ In the cross-linguistic literature, non-tensed clauses have been claimed to be structurally smaller than tensed clauses (e.g. Ramchand and Svenonius 2014; Lohninger and Wurmbrand 2020; Wurmbrand and Lohninger 2023). Once we assume that subjunctive morphology *-j* is introduced by a Mood head, and (past) tense is introduced by a T head, we can state the following:

- (55) T > Mood

Thus, only tensed clauses need to have a TP layer in order to have [+tense] encoded, subjunctives only need to have a MoodP layer. Lohninger and Wurmbrand (2020) and Wurmbrand and Lohninger (2023) claim something similar by saying that subjunctives tend to have a reduced structure across languages. I go further by explicitly proposing that subjunctives are not only internally reduced in size: since there is no CP layer in Hungarian, the subjunctives’ lack of TP layer makes them externally smaller; they are minimally MoodPs, not TPs. Cross-linguistically, non-tensed clauses have been found to be more transparent than tensed clauses (e.g. Wurmbrand 2015). The most well-known example of this is the ban on *hyperraising* in English (Chomsky 1973), discussed in the previous section.

I argue that the size-difference between subjunctive and tensed clauses explains the (im)possibility of particle extraction out of them. More specifically, the ban on Hungarian

¹¹ In fact, with subjunctive-selecting verbs like *akar* ‘want’, subjunctive clauses obligatorily replace infinitives when the subject of the embedded clause is different from the subject of the matrix clause.

particle movement from tensed clauses falls under the Williams Cycle in (36) if Part is sandwiched between T and Mood in *fseq*.¹²

(56) T>Part>Mood

That the Part head is below the head responsible for encoding tense is supported by the observation that particles are involved in encoding aspect and telicity (e.g. É. Kiss 2006; Hegedűs 2020), and heads associated with aspect are assumed to be lower than heads associated with tense (e.g. Ramchand and Svenonius 2014).¹³

In sum, the strict Williams Cycle (repeated below) with the right assumptions covers the opacity-difference between subjunctive and tensed clauses. Particle extraction to matrix Spec,PartP is possible from within at least some of the subjunctive clauses because these are MoodPs and Mood is lower in *fseq* than Part. On the other hand, tensed clauses are minimally TPs, and since T>Part in *fseq*, particle movement is impossible from tensed clauses.¹⁴

(57) **Strict Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, if $\alpha \geq \beta$ in some functional sequence.

The reference to *minimal sizes* leaves open the possibility that subjunctives take particles, foci, sentential adverbs, and relative pronouns whereby they grow to the size of PartP, FocP, Comp, and RelP. The locality of selection should not be a problem in such cases assuming that the tense or subjunctive features can be percolated to the maximal projection of the embedded clause (as in Grimshaw 2000).

¹²The generalization also holds if one handles subjunctive as some kind of future aspect and assumes that *fseq*=T>Part>Asp. Wurmbrand (2015) has discussed cases in which future clauses can be assumed to be smaller and more transparent to long-movement.

¹³Notice that the prohibition against particle-extraction out of TPs falls under the strict version of the Williams Cycle in (36) even if particle movement is assumed to land in Spec,TP (as in e.g. Surányi 2009).

¹⁴In this paper, I do not discuss infinitival complements. Similar to Hungarian subjunctives, Hungarian infinitival clauses are not tensed, either. With Bartos (2000) and É. Kiss (2002), I assume that the *-ni* suffix heads the InfP. If this suffix is part of the clausal *fseq*, then T>Inf. Furthermore, an anonymous reviewer mentions that there are certain verbs like *parancsol* ‘command’ and *sikerül* ‘become achieved’, whose subjunctive complement clauses are opaque to particle-extraction. These complement clauses are possibly larger than other subjunctives. However, this does not influence my central claim: While particle extraction is *never* possible from tensed clauses, it *can* be grammatical from subjunctives.

This section demonstrated that although clause-types do not show differences in their word order and left-periphery, the degree to which they are opaque can be analyzed as a consequence of their size-differences. Informally, the occurrence of a focus or a particle in the left periphery has the same particle-movement-blocking effect as the switch between subjunctive and tensed clauses.

5 Conclusion and Williams Cycle beyond movement

In this paper, I showed that the opacity of Hungarian embedded clauses patterns with a stricter version of the Williams Cycle.

- (58) **Strict Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, if $\alpha \geq \beta$ in some functional sequence.

If $fseq = Rel > Com > Foc2 > Foc1 > T > Part > Mood > V$, data summarized in (59) falls under the Williams Cycle in (58): Since particle extraction lands in one of the lowest phrases, it is only possible from the smallest subjunctive clauses, which I referred to as MoodPs. Focus extraction lands in a higher position (Spec,Foc2P); correspondingly, it is possible from clauses up to the size of Foc1P. Finally, since relative movement lands in the highest position in the clause (Spec,RelP), it can proceed even from ComPs.

- (59) **Embedded-tense projections leading to the same effect as overt left-periphery positions**

	Particle extraction	Focus extraction	Relative extraction
Embedded MoodP	✓	✓	✓
Embedded PartP	×	✓	✓
Embedded TP	×	✓	✓
Embedded Foc1P	×	✓	✓
Embedded ComP	×	×	✓

On the one hand, size-related bounding may have an overt reflection. For instance, if a clause has a focus, it will be minimally a Foc1P. In this case, it is opaque to movement

types landing in projections whose head is lower in *fseq* than the Foc1 head. In contrast, the size-differences between clause types (subjunctive vs. tensed clauses) is not reflected by overt left-periphery material. However, once we propose that these clause types have different minimal sizes (e.g. only tensed clauses need a T head), the opacity-differences between clause types can be derived from their size.

This paper limited its attention to syntactic movement; however, other syntactic dependencies may also exhibit Williams Cycle effects, e.g. Poole (2023) discusses Williams Cycle for case. In Hungarian, one phenomenon possibly exhibiting Williams Cycle effects beyond movement is definiteness agreement. Throughout the paper, object clauses (or their clausal expletives bearing accusative case) triggered definite agreement on the matrix verb, cf. e.g. (47) and (48).¹⁵ In contrast, object *wh*-pronouns bearing accusative case obligatorily trigger indefinite agreement on the verb within clauses:

- (60) $[_{FOC} \text{ Mi-}t]_2 \quad \text{vet-}t\{-ek/ \quad *{-ék}\}_1 \quad [_{PART} \text{ meg}] \text{ } __1 \text{ } __2?$
 what-ACC buy-PST-3PL.INDF -3PL.DEF PART
 ‘What did they buy?’

Moreover, it has been observed that the long movement of *wh*-objects bearing accusative to matrix Spec,FocP changes the matrix agreement from definite to indefinite as in (61) (e.g. É. Kiss 1987).

- (61) $[_{FOC} \text{ Mi-}t]_2 \quad \text{lát-}t\{-ál/ \quad *{-ad}\}_1 \quad __1, \quad [\text{ hogy } [_{PART} \text{ meg}]$
 what-ACC see-PST-2SG.INDF -2SG.DEF that PART
 $\text{vet-}t-ek \quad __2]?$
 buy-PST-3PL.INDF
 ‘What did you see them buying?’

Relative pronouns also trigger indefinite agreement on the verb of the clause they are base generated in:

¹⁵ Whether every object clause comes with an expletive is an open question in the literature (cf. Den Dikken 2018). Nevertheless, the central claim is not influenced by this analytical choice; all that we are interested in is how the default definite agreement is influenced by extracted accusative arguments.

- (62) *Ez az a kocsi, [REL ami-t]₁ [PART meg] vet-t{-ek/ *-ék} ____1.*
 this that the car which-ACC PART buy-PST-3PL.INDF -3PL.DEF
 ‘This is the car that they bought.’

To my knowledge, it has not been noticed that unlike long-moving *wh*-phrases in (61), long-moving relative pronouns are compatible with both definite and indefinite agreement in the clause in which they land:

- (63) *Ez az a kocsi, [REL ami-t]₁ lát-t{-ad/ -ál}, [hogy [PART*
 this that the car which-ACC see-PST-2SG.DEF -2SG.INDF that
meg]₂ vet-t-ek ____1 ____2]
 PART buy-PST-3PL.INDF
 ‘This is the car that you saw that they bought.’

The contrast between the agreement in (61) and (63) may have a Williams Cycle-based explanation on the assumption that *ComP* may be projected in the complement clause of verbs like *lát* ‘see’ even in the absence of an overt sentence adverb (see Footnote 3). As argued above, focus movement cannot proceed from larger clauses than *Foc1P*, whereas relative movement is less constrained and can proceed from clauses as large as *ComP*. That is, while both focusing and relativization can proceed from *MoodP*, *PartP*, *TP*, and *Foc1P*, larger clauses like *ComPs* and *Foc2Ps* are only transparent to relativization. It is possible that object agreement has the same size-sensitivity as long focus movement, that is, whenever long focus movement can proceed, object agreement with the moving focus is obligatory as in (61). In contrast, indefinite object agreement only co-occurs with relativization if relativization proceeds from sufficiently small clauses (e.g. *MoodPs* and *TPs*); whenever relativization proceeds from some larger clause (e.g. *ComP*), the matrix verb cannot exhibit indefinite agreement with the moving relative pronoun. Since by assumption *ComP* may or may not be projected in the embedded clause in (63), here object agreement with the matrix verb is optional.

A more specific hypothesis may be drawn from the landing site of focus/relative movement and the landing site of verbs. Assume that even if the object-agreement probe

located on a relatively low head like *v*, it can search and agree as long as the verb moves and pied-pipes it. In clauses without foci, the verb moves up to Mood (subjunctive clauses) or T (tensed clauses), that is, the object agreement probes can probe while MoodP and TP are being processed, but not after that. In clauses with foci, the verb moves up to Foc1 or Foc2, so probing is possible while Foc1P or Foc2P is being processed.

In the spirit of Keine (2016; 2019; 2020), agreement could be constrained in a Williams Cycle-like fashion:

- (64) **Williams Cycle for agreement:** A probe on β cannot search in α P, if $\alpha > \beta$ in some functional sequence.

By (64), an object agreement probe landing on the matrix Mood, T, Foc1, or Foc2 cannot probe into an embedded ComPs because Mood, T, Foc1, or Foc2 are all lower than Com in the *fseq*. In contrast, ComPs are transparent to relative movement because Rel > Com in the *fseq*. Thus, while in cases where the embedded ComP is not projected the embedded MoodP/TP/Foc1P/Foc2P is transparent to both relative movement and object agreement, in other cases, the embedded clause is a ComP, which is transparent to relative extraction but opaque to object agreement. In short, the optional presence of ComP derives the optionality of matrix agreement with the object relative pronoun.

Focus movement is driven by a probe on Foc1 or Foc2, which is also the landing site of the verb containing the object agreement probe in clauses with foci. Since the two kinds of probes are located at the same height in the extended projection, whenever the focusing probe on Foc constrained by (58) successfully finds a focused object in the embedded clause, the object agreement probe constrained by (64) successfully agrees with the focus.¹⁶

While the paper provided an extensive discussion for the Williams Cycle effects involving cross-clausal movement, I leave the detailed discussion of agreement and other

¹⁶ A theory explaining Williams Cycle effects in Hungarian needs to explain why extraction is subject to the strict Williams Cycle in (58) whereas agreement is subject to the normal/weak Williams Cycle in (64). While it is not uncommon that agreement exhibits looser locality restrictions than movement (e.g. Polinsky and Potsdam 2001), I leave this question for further study. As Footnote 8 mentioned, the weak Williams Cycle is compatible with the Hungarian data, however, the main text used the strict version as it can cover more of the Hungarian data in the paper.

syntactic operations to further research.

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Bibliography

- Abels, Klaus. 2012. The Italian Left Periphery: A View from Locality. *Linguistic Inquiry* 43:229–254.
- Allwood, Jens. 1982. The complex NP constraint in Swedish. In *Readings on unbounded dependencies in Scandinavian languages*, eds. Elisabet Engdahl and Eva Ejerhe, 15–32. Stockholm: Almqvist and Wiksell International.
- Bartos, Huba. 2000. Morphosyntax and interpretation: The syntactic aspects of Hungarian inflectional phenomena. PhD Thesis, Eötvös Loránd University, Budapest.
- Chomsky, Noam. 1973. Conditions on transformations. In *A Festschrift for Morris Halle*, eds. Stephen Anderson and Paul Kiparsky, 232–286. New York: Holt, Rinehart and Winston.
- Chomsky, Noam. 1986. *Barriers*. Linguistic Inquiry Monographs 13. Cambridge, Mass: The MIT Press.
- Csirmaz, Anikó. 2004. Particles and phonologically defective predicates. In *Verb Clusters: A study of Hungarian, German and Dutch*, eds. Katalin É Kiss and Henk van Riemsdijk, *Linguistik Aktuell/Linguistics Today* 69, 225–252. Amsterdam: John Benjamins Publishing Company.

- Deal, Amy Rose. 2020. *A theory of indexical shift: meaning, grammar, and crosslinguistic variation*. Linguistic Inquiry Monographs 82. Cambridge, Massachusetts: The MIT Press.
- Den Dikken, Marcel. 2018. *Dependency and Directionality*. Cambridge Studies in Linguistics. Cambridge: Cambridge University Press.
- Dékány, Éva, and Veronika Hegedűs. 2015. Word order variation in Hungarian PPs. In *Approaches to Hungarian 14*, eds. Katalin É. Kiss, Balázs Surányi, and Éva Dékány, 95–120. Amsterdam: John Benjamins Publishing Company.
- É. Kiss, Katalin. 1987. *Configurationality in Hungarian*. Studies in Natural Language and Linguistic Theory 3. Dordrecht: Springer.
- É. Kiss, Katalin. 1994. Sentence structure and word order. In *The Syntactic Structure of Hungarian*, eds. Ferenc Kiefer and Katalin É. Kiss, 1–90. San Diego, California: Academic Press.
- É. Kiss, Katalin. 1998a. Multiple Topic, One Focus? *Acta Linguistica Hungarica* 45:3–29.
- É. Kiss, Katalin. 1998b. Verbal Prefixes or Postpositions? Postpositional Aspectualizers in Hungarian. In *Approaches to Hungarian 6.: Papers from the Amsterdam Conference*, eds. de Groot Casper and István Kenesei, 123–148. Szeged: József Attila Tudományegyetem (JATE).
- É. Kiss, Katalin. 2002. *The Syntax of Hungarian*. Cambridge Syntax Guides. Cambridge: Cambridge University Press.
- É. Kiss, Katalin. 2006. *Event Structure And The Left Periphery: Studies on Hungarian*. Studies in Natural Language and Linguistic Theory 68. Dordrecht: Springer.
- É. Kiss, Katalin. 2010. An adjunction analysis of quantifiers and adverbials in the Hungarian sentence. *Lingua* 120:506–526.
- Egedi, Barbara. 2021. PPs as adjuncts. In *Syntax of Hungarian: Postpositions and Postpositional Phrases*, eds. Katalin É. Kiss and Veronika Hegedűs, 371–430. Amsterdam University Press.
- Egressy, János. 2023. Size- and position-dependent opacity in Hungarian. Ph.D. dissertation, UCLA.
- Farkas, Donka F., and Jerrold M. Sadock. 1989. Preverb Climbing in Hungarian. *Language*

65:318–338.

- Georgi, Doreen. 2014. Opaque interactions of Merge and Agree: On the nature and order of elementary operations. PhD Thesis, Universität Leipzig.
- Georgi, Doreen. 2017. Patterns of Movement Reflexes as the Result of the Order of Merge and Agree. *Linguistic Inquiry* 48:585–626.
- Grimshaw, Jane. 2000. Locality and Extended Projection. In *Lexical insertion and specification*, eds. Peter Coopmans, Martin B.H. Everaert, and Jane Grimshaw, Current Issues in Linguistic Theory 197, 115–134. Amsterdam: John Benjamins Publishing Company.
- Halm, Tamás. 2021. Radically Truncated Clauses in Hungarian and Beyond: Evidence for the Fine Structure of the Minimal VP. *Syntax* 24:376–416.
- Halpert, Claire. 2019. Raising, unphased. *Natural Language & Linguistic Theory* 37:123–165.
- Hegedűs, Veronika. 2020. A névutón át, közel az igekötőhöz: A grammatikalizálódó ragvonzó névutók kategóriájáról [Through the postposition, near the verbal particle: On the category of inflection selecting postpositions]. *Általános Nyelvészeti Tanulmányok* 32:83–99.
- Horvath, Julia. 1981. Aspects of Hungarian Syntax and the Theory of Grammar. Ph.D. dissertation, University of California, Los Angeles, Los Angeles, CA.
- Horvath, Julia. 1986. *FOCUS in the theory of grammar and the syntax of Hungarian*. Studies in Generative Grammar 24. Dordrecht: Foris.
- Keine, Stefan. 2016. Probes and their Horizons. Ph.D. dissertation, University of Massachusetts Amherst, Amherst, MA.
- Keine, Stefan. 2019. Selective Opacity. *Linguistic Inquiry* 50:13–62.
- Keine, Stefan. 2020. *Probes and their horizons*. Cambridge, MA: The MIT Press.
- Koopman, Hilda Judith, and Anna Szabolcsi. 2000. *Verbal Complexes*. Cambridge, MA: The MIT Press.
- Krifka, Manfred. 2023. Layers of assertive clauses: propositions, judgements, commitments, acts. In *Propositional Arguemnts in Cross-Linguistic Research: Theoretical and Empirical Issues*, eds. Jutta M Hartman and Angelika Wöllstein, 115–181. Tübingen: Narr Verlag.
- Lohninger, Magdalena, and Susi Wurmbrand. 2020. Typology of complement clauses. In *Handbook of Clausal Embedding*, eds. Anton Benz, Werner Frey, Hans-Martin Gärtner,

- Manfred Krifka, Mathias Schenner, and Marzena Żygis. Oxford: Oxford University Press.
- Manetta, Emily. 2006. Peripheries in Kashmiri and Hindi-Urdu. Ph.D. dissertation, University of California, Santa Cruz, Santa Cruz, California.
- Manetta, Emily. 2011. *Peripheries in Kashmiri and Hindi-Urdu*. Amsterdam: John Benjamins Publishing Company.
- Müller, Gereon. 2014a. A local approach to the Williams Cycle. *Lingua* 140:117–136.
- Müller, Gereon. 2014b. *Syntactic buffers: A local-derivational approach to improper movement, remnant movement, and resumptive movement*. Linguistische Arbeitsberichte 91. Leipzig: Institut für Linguistik.
- Olsvay, Csaba. 2000. Formális jegyek egyeztetése a magyar nemsemleges mondatokban. In *A mai magyar nyelv leírásának újabb módszerei*, eds. László Büky and Márta Maleczki, IV, 191–151. Szeged: Szeged University Press.
- Polinsky, Maria, and Eric Potsdam. 2001. Long-Distance Agreement And Topic In Tsez. *Natural Language & Linguistic Theory* 19:583–646.
- Poole, Ethan. 2023. Improper case. *Natural Language & Linguistic Theory* 41:347–397.
- Puskás, Genoveva. 2000. *Word Order in Hungarian: The syntax of $\bar{A}$ -positions*. Linguistik Aktuell/Linguistics Today 33. Amsterdam: John Benjamins Publishing Company.
- Ramchand, Gillian, and Peter Svenonius. 2014. Deriving the functional hierarchy. *Language Sciences* 46:152–174.
- Rizzi, Luigi. 1990. *Relativized minimality*. Cambridge, MA: The MIT Press.
- Rizzi, Luigi. 1997. The Fine Structure of the Left Periphery. In *Elements of Grammar: Handbook in Generative Syntax*, ed. Liliane Haegeman, Kluwer International Handbooks of Linguistics, 281–337. Dordrecht: Springer.
- Starke, Michal. 2001. Move Dissolves into Merge: a Theory of Locality. Ph.D. dissertation, University of Geneva, Geneva.
- Surányi, Balázs. 2002. Multiple Operator Movements in Hungarian. PhD Thesis, Utrecht University, Utrecht, The Netherlands.
- Surányi, Balázs. 2008. Határozószók és mondattartományok [Adverbs and sentence domains]. *Nyelvtudományi Közlemények* 105:163–192.

- Surányi, Balázs. 2009. Verbal particles inside and outside vP. *Acta Linguistica Hungarica* 56:201–249.
- Surányi, Balázs. 2012. Interface configurations: identificational focus and the flexibility of syntax. In *Contrasts and Positions in Information Structure*, eds. Ad Neeleman and Ivona Kučerová, 87–101. Cambridge: Cambridge University Press.
- Szendrői, Kriszta. 2003. A stress-based approach to the syntax of Hungarian focus. *The Linguistic Review* 20 37–78.
- Williams, Edwin. 2003. *Representation Theory*. Cambridge, MA: The MIT Press.
- Williams, Edwin. 2013. Generative Semantics, Generative Morphosyntax. *Syntax* 16:77–108.
- Williams, Edwin Samuel. 1974. Rule ordering in syntax. PhD Thesis, Massachusetts Institute of Technology.
- Wurmbrand, Susi. 2015. Restructuring cross-linguistically. *Proceedings of the North Eastern Linguistics Society Annual Meeting* 45:227–240.
- Wurmbrand, Susi, and Magdalena Lohninger. 2023. An implicational universal in complementation—Theoretical insights and empirical progress. In *Propositional arguments in cross-linguistic research: Theoretical and empirical issues*, eds. Jutta M. Hartmann and Angelika Wöllstein, 183–232. Tübingen: Narr Verlag.

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